

Solution Requirements

Date	1 July 2026
Team ID	—
Project Name	India's Agriculture Crop Production Analysis
Maximum Marks	4 Marks

Solution Requirements Overview

Define Functional and Non-Functional Requirements to map exactly what your solution will do and how well it will perform. Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do), while Non-Functional Requirements (NFR) specify operational constraints, properties, and standards judging the overall execution behavior (How the system should be).

Step 1: Functional Requirements (FR)

S.NO	REQUIREMENT CATEGORY	REQUIREMENT DESCRIPTION	PRIORITY
1	Data Collection & Integration	Collect and integrate crop production data from multiple sources such as government datasets (data.gov.in), Ministry of Agriculture APIs, and weather services.	High
2	Data Management	Store and manage structured data including crop type, region, season, area harvested, production quantity, yield, and year.	High
3	Data Analysis	Perform analysis on crop production trends, seasonal variations, and regional performance using statistical methods and predictive models.	High
4	Visualization & Dashboards	Provide interactive dashboards with charts, graphs, and maps to visualize crop production trends, state-wise comparisons, and year-on-year performance.	High
5	Geospatial Analysis	Use GIS mapping to display crop production across Indian states and districts, highlighting high and low production zones.	Medium
6	Reports & Export	Generate and export reports (PDF/Excel/CSV) for crop production statistics, insights, and comparisons.	Medium
7	User Management	Manage user roles (Admin, Analyst, Viewer) with secure login and role-based access to features and data.	Medium



S.NO	REQUIREMENT CATEGORY	REQUIREMENT DESCRIPTION	PRIORITY
6	 Business Rules, etc.	The platform must analyze crop production data across selected time periods (weekly, monthly, annual), compute yield and production metrics using standardized agronomic formulas, and flag records with missing critical parameters (e.g., area, yield, crop type) as incomplete.	High
7	 Compliance to Laws	The platform ensures complete data transparency and adheres strictly to the Right to Information (RTI) guidelines, MeitY data policies, and the Digital Personal Data Protection Act, 2023. All user and data access activities are logged with traceable audit trails.	High
8	 Other	Includes an optional field-level data quality checker and an automated outlier detection engine to validate abnormal values in crop yield and production before generating analytics and reports.	Medium

Step 2: Non-Functional Requirements (NFR)

S.NO	NFR CATEGORY	REQUIREMENT DESCRIPTION	TARGET METRIC / ACCEPTANCE CRITERIA
1	 Performance & Speed	Data ingestion, preprocessing, analytics computation, and dashboard rendering must deliver results swiftly for large-scale agricultural datasets.	Dashboard load time and report generation completes in < 5 seconds for standard queries.
2	 Scalability	The architecture uses cloud-native microservices capable of handling growth in data volume, users, and concurrent analytical requests.	Horizontal scaling reliably supports up to 10,000 concurrent users and TB-scale agricultural datasets.
3	 Security & Data Privacy	All stored data, user credentials, and analytics results must be secured with encryption and access controls to protect sensitive agricultural information.	End-to-end data encryption (AES-256), secure APIs, and role-based access control implementation.
4	 Reliability & Availability	The platform must maintain high availability and ensure continuous service during data processing and reporting operations.	Guarantees 99.9% uptime monthly with automated backup and disaster recovery.
5	 Usability & Accessibility	The platform should offer intuitive dashboards, mobile responsiveness, and accessibility features for all user groups.	Complies with WCAG 2.1 AA standards and supports mobile and regional language interfaces.
6	 Other	The codebase must follow modular design principles with clean documentation and automated testing for long-term maintainability.	Achieves > 90% code coverage with automated tests and continuous integration pipelines.